Djordje Miladinovic

Curriculum Vitæ

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Summary

- **Profile**: AI Director | PhD in computer science (AI/ML) | Swiss resident | Serbian citizen.
  13+ years in AI/ML research | 18+ years in software development | 7+ years in project leading | 5+ years in life sciences.
- **Work**: Designing and leading the end-to-end development of scalable AI/ML and software products, from concept to deployment.
- **Mission**: My goal is to utilize technology, innovative design, and data-driven insights to improve individual and societal well-being, fuel business growth, and deliver best-in-class solutions with measurable impact.

Experience

AI Director, *AI department of GSK plc* ***Aug'26-Present***

- Leading development of AI agents for continual scientific discovery.

Senior AI/ML Engineer, *Reality Labs* ***Sep'25-Aug'26***

- Developing elements of an agentic AI framework that powers Meta's wearable devices, enabling immersive, intelligent experiences for millions of users worldwide.
  Tech: Agentic AI | Programmatic tool calling | Real-time inference | Few-shot learning;

Senior AI/ML Engineer, *AI department of GSK plc* ***Oct'21-Sep'25***

- Led the development of a foundational model integrating multimodal omics data to facilitate preclinical stage of drug discovery – *adopted by three R&D units at GSK, enhancing decision-making in target identification.*
  Tech: Neural networks | Representation learning | Large scale data pipelines | Active learning | Counterfactuals;
- Supported clinical efforts to optimize the use of existing GSK drugs.
  Tech: Classic ML | Survival analysis | Random forests | Data from electronic health records (EHRs);

Machine Learning Researcher, *ETH Zürich* ***Feb'17-Jun'21***

- See the full list of my publications at Google Scholar. Notable projects include:
- SDN neural networks: Co-developed a neural architecture for realistic image synthesis – *state-of-the-art performance.*
  Tech: Generative image modeling | VAE | LSTM | CNN | Autoregressive models | Representation learning;
- Sleep-learning web platform: Led a team of researchers to develop a web-based large-scale machine learning platform for sleep pattern recognition from EEG/EMG signals – *now commercialized with over 100K jobs processed.*
  Tech: CNN | Fourier transform | Image classification | Web service;

Machine Learning Researcher (Visitor), *Max Planck Institute for Intelligent Systems* ***Sep'18-Dec'18***

- Co-developed a framework for evaluating disentangled representations.
  Tech: Representation learning | Causal analysis | Robust machine learning;

Machine Learning Researcher & Engineer (Intern), *Logitech Europe S.A.* ***Sep'16-Feb'17***

- Developed an AI/ML product to predict potential premium users and reduce churn rate.
  Tech: Regression | Sentiment analysis | Random forests | SQL;

Machine Learning Researcher (Visitor), *Disney Research Studios, Zurich* ***Jan'16-Sep'16***

- Developed a software platform for detecting mechanical failures in Disney's humanoid robots based on IMU sensor readings. The platform uses novel machine learning algorithm to asses robot degradation in a human-like way.
  Tech: Kalman filter | Motion capture | IMUs | Dynamic time warping | Time series classification | 3D rendering;

Education

PhD in Computer Science – Machine Learning, *ETH Zürich* ***Sep'17-Jun'21***

- *Thesis:* "On training Deep Generative Models with Latent Variables" – Prof. Joachim M. Buhmann.
- *Research areas:* Generative image and text modeling | Variational autoencoders | Representation learning | Deep learning | Large language models | Machine learning for biology;

MS in Computer Science – Machine Learning, *ETH Zürich* ***Sep'13-Jun'16***

- *Thesis:* "Perceptual Analysis Framework for Discovering Anomalies in Humanoid Motions" – Prof. Otmar Hilliges.
- *Focus:* Machine learning | Software engineering | Operating systems | Distributed computing;

BS in Computer Science – Software Engineering, *ETF Belgrade* ***Sep'09-Sep'13***

- *Thesis:* "Java-based Interactive Software Simulator of a CISC Processor" – Prof. Zaharije Radivojević.
- *Focus:* Software engineering and design | Algorithms and data structures | Operating systems | Hardware design;

Technical Skills

Programming	*Advanced:* Python (PyTorch);
	Good but rusty: C/C++ | Java | Javascript | Bash | SQL;
Infrastructure	*Data formats:* Parquet | H5 | H5AD | CSV | YAML | JSON | Pickle;
(Varying levels)	*Databases:* PostgreSQL | MongoDB | SQLite | Milvus;
	Big data: Spark | Hive | Hadoop;
	AI/ML frameworks: Pytorch Lightning | Ray | Weights & Biases | MLflow | HuggingFace | Kubeflow;
	High Performance Computing (HPC): SLURM | LSF | Condor;
	Middleware: Celery | Redis | Kafka;
	Cloud: GCP/GCS | AWS/S3 | Microsoft Azure;
	Deployment: Docker | Kubernetes;
	CI/CD: GitHub Actions | CircleCI;
	Web: Node.js | Django | HTML/CSS;
Python libraries	*Data Science:* Pandas | SciPy | Seaborn | NumPy | Scanpy | Scrapy | OpenCV;
	AI/ML: PyTorch | Scikit-learn | SpaCy | Fairseq | LangChain | Tensorflow;
Design tools	Figma | Inkscape | Blender | PowerPoint | Adobe Illustrator;

Languages

Serbian (Native) | English (Fluent) | German (Intermediate) | Spanish (Beginner);

Hobbies

Waterpolo | Swimming | Basketball | Tennis | Chess | Cinematography | Reading;